

cloud deployment models:-

* There are four main types of cloud deployment models:-

① Private

② Public

③ Hybrid

④ Community

① Public cloud:-

* Public cloud is a cloud computing model where cloud services are provided over the internet by a third-party cloud service provider.

* services are available to the general public on a pay-as-you-go basis.

Ex:- ~~those~~
characteristics:-

① Scalability:-

Public cloud offer vast resources that can scale up (or) down based on demand.

Advantages:-

* Low cost
* No maintenance

* can be access from anywhere
* High scalable

Dis
* Less customizable
* Less secure

② cost-effective:-

user only pay for the resources they consume, reducing upfront infrastructure costs.

③ shared Resources:-

Resources are shared among multiple users leading to cost savings but potentially impacting performance during peak times.

④ Accessibility:-

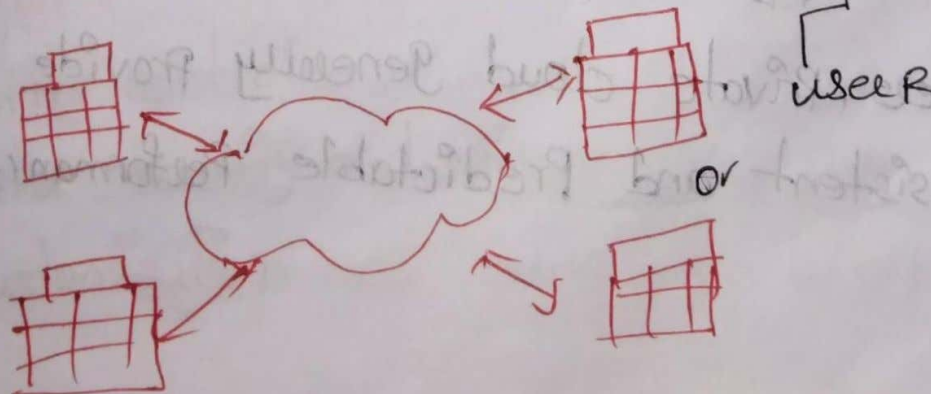
Public clouds can be accessed from anywhere with an internet connection.

use cases:-

- * software as a service (saas) applications.
- * web hosting and email services
- * collaborative tools and productivity suite

Examples Providers:-

AWS, Microsoft Azure, Google cloud, IBM



② Private cloud :-

Private cloud is a cloud computing model where cloud services are dedicated to a single organization and are not accessible to the general public.

* So it is also known as corporate cloud (or) internal cloud.
* "Internal cloud" means that it allows the accessibility of systems and services within a specific boundary or organization.

① control and security :-

Private cloud offer ~~ge~~ greater control over data and security, making them suitable for highly sensitive data and compliance requirements.

② customization :-

Organizations can tailor the private cloud to meet their specific needs and infrastructure.

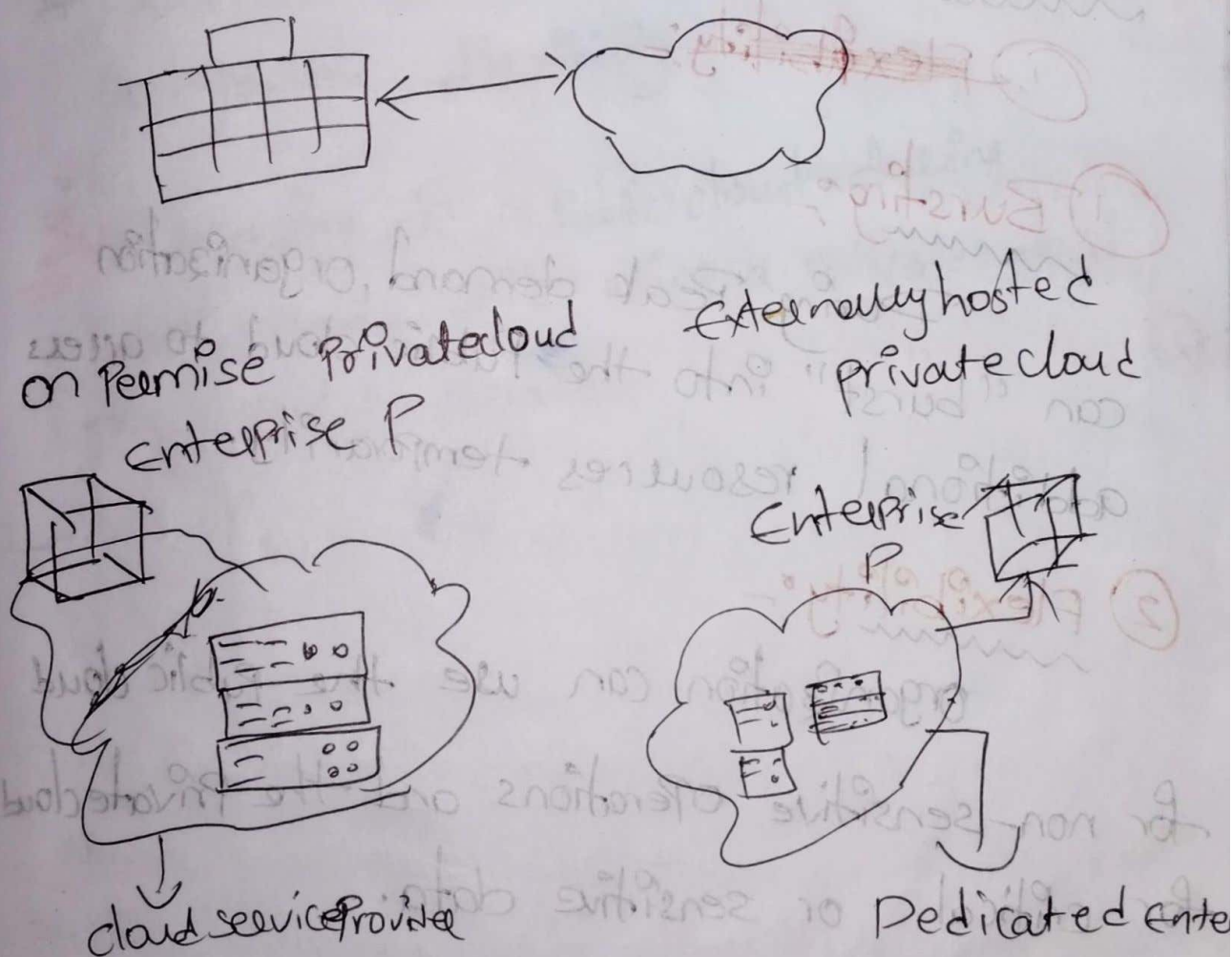
③ Performance :-

As resource are not shared with others, private cloud generally provide more consistent and predictable performance.

Use case:-

- * Government agencies handling classified data.
- * Financial institutions with strict security and regulatory requirements.
- * Large enterprises with significant internal IT needs.

Ex:- Openstack, VMware



Advantage:-

- ① Better control
- ② Data security and Privacy
- ③ customization

Disadvantage

- ① less scalable
- ② costly.

③ Hybrid cloud :-

Hybrid cloud is a cloud computing model that combines both Public and Private cloud deployments.

* It allows seamless integration and data flow between the two environments.

characteristics :-

① ~~Flexibility~~ :-

① Bursting :-

During Peak demand, organization can "burst" into the Public cloud to access additional resources temporarily.

② Flexibility :-

organization can use the Public cloud for non-sensitive operations and the Private cloud for critical or sensitive data.

③ Data Backup and Redundancy :-

Hybrid clouds offer a redundant backup options, ensuring data available even if one cloud experiences an outage.

Use-cases:-

* E-commerce companies that experience fluctuating traffic.

* Healthcare organizations needing to comply with regulation while leveraging public cloud for less sensitive data.

* Enterprises seeking cost optimization and greater flexibility.

Ex:- Bursting to a public cloud during peak loads while using a private cloud for regular operations.

community cloud:-

community cloud is a cloud computing model where cloud resources are shared among several organizations with common ~~same~~ interests or objectives.

characteristics:-

① limited access:-

Access to the community cloud is restricted to specific organizations with shared purposes.

② cost sharing:-

Participants share the costs the community cloud, making it more cost-effective.

③ security and Privacy:-

community cloud are designed to address the specific security and compliance needs of the participating organizations.

Use Cases:-

① Research consortiums collaborating on data-intensive projects.

② Healthcare organization sharing patient data while ~~maintaining~~ maintaining privacy compliances.

* Educational institutions with shared resources for E-learning Platforms.

Ex:- Government agencies, research institutions.

community cloud is similar as private cloud, the only difference is the set of users.

* In Private cloud only one company own the server, in the case of a community cloud several organizations with similar backgrounds share the infrastructure and related resources.



community cloud is best for joint Business organization
Research organization etc---

Adv

- * cost effective
- * Flexible and scalable
- * security